

Krishnendu Chakrabarty

professor

基本信息

姓名: Krishnendu Chakrabarty

所属院校: 杜克大学

地区: United States

学科领域

电子与计算机工程

生物医学工程

集成电路设计与集成系统

人工智能

微电子科学与工程

研究领域

生物医学传感与检测技术

集成电路设计与制造

计算机体系结构

嵌入式系统与硬件设计

生物信号处理与模式识别

EDA工具与电路验证

电路与系统设计

数字集成电路设计

深度学习与神经网络

个人简介

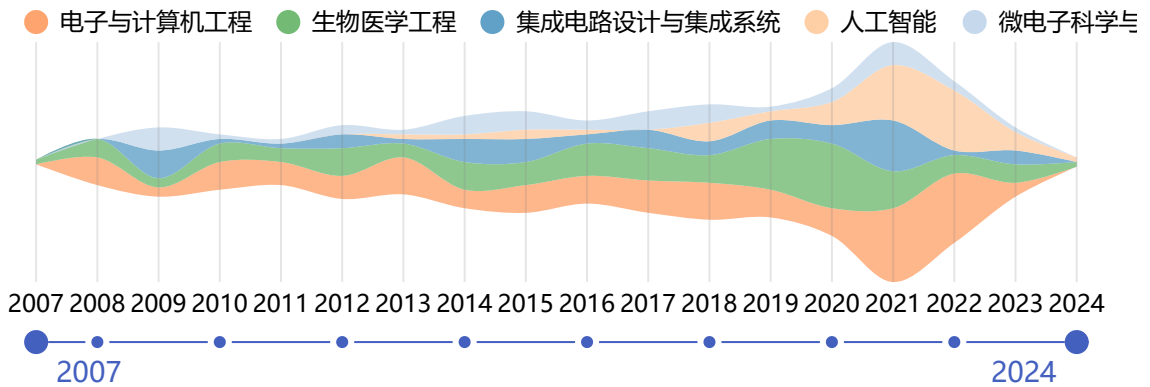
Krishnendu Chakrabarty is the Fulton Professor of Microelectronics at the School of Electrical, Computer, and Energy Engineering at Arizona State University. His research focuses on electronic design automation, testing and design-for-testability, microfluidics, computer engineering, and sensor networks. He has been widely cited, with over 39,417 citations to his work since 2020, and holds an h-index of 94. Chakrabarty collaborates with numerous esteemed co-authors across various institutions and industries.

工作经历

Indian Institute of Technology Kharagpur - Distinguished Visiting Professor

New York University - Professor	-
Intel Corp	-
imec - Scientific Director	-
Florida International University - Professor	-
Washington State University	-
Facebook	-
Intel Labs	-
Apple Inc.	-
University of Florida - Professor	-
Karlsruhe Institute of Technology - Professor	-
Duke University - Professor	-
Baebies Inc.	-
University of - Professor	-
NVIDIA Corp.	-
University of Ioannina	-
Duke University	-
National Chiao Tung University - Professor	-
University of Michigan - Professor	-
School of Electrical and Computer and Energy Engineering - Professor	-

近年发表的论文



Dynamic Adaptation Using Deep Reinforcement Learning for Digital Microfluidic Biochips

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Liang, Tung-Che; Chang, Yi-Chen; Zhong, Zhanwei; Bigdeli, Yaas; Ho, Tsung-Yi; Chakrabarty, Krishnendu; Fair, Richard

被引用次数: 0

2024

Enhanced Built-In Self-Diagnosis and Self-Repair Techniques for Daisy-Chain Design in MEDA Digital Microfluidic Biochips

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Ling; Li, Zipeng; Huang, Xing; Chakrabarty, Krishnendu

被引用次数: 2

2023

Fusion of IoT, AI, Edge-Fog-Cloud, and Blockchain: Challenges, Solutions, and a Case Study in Healthcare and Medicine

IEEE INTERNET OF THINGS JOURNAL

Firouzi, Farshad; Jiang, Shiyi; Chakrabarty, Krishnendu; Farahani, Bahar; Daneshmand, Mahmoud; Song, Jaeseung; Mankodiya, Kunal

被引用次数: 31

2023

Stool Image Analysis for Digital Health Monitoring By Smart Toilets

IEEE INTERNET OF THINGS JOURNAL

Zhou, Jin; McNabb, Jackson; DeCapite, Nick; Ruiz, Jose R.; Fisher, Deborah A.; Grego, Sonia; Chakrabarty, Krishnendu

被引用次数: 6

2023

Accelerating Graph Neural Network Training on ReRAM-Based PIM Architectures via Graph and Model Pruning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ogbogu, Chukwufumnanya O.; Arka, Aqeeb Iqbal; Pfromm, Lukas; Joardar, Biresh Kumar; Doppa, Janardhan Rao; Chakrabarty, Krishnendu; Pande, Partha Pratim

被引用次数: 5

2023

Machine Learning-Based Rowhammer Mitigation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Joardar, Biresh Kumar; Bletsch, Tyler K.; Chakrabarty, Krishnendu

Deep Reinforcement Learning-Based Approach for Efficient and Reliable Droplet Routing on MEDA Biochips

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Elfar, Mahmoud; Chang, Yi-Chen; Ku, Harrison Hao-Yu; Liang, Tung-Che; Chakrabarty, Krishnendu; Pajic, Miroslav

被引用次数: 1

2023

Learning Malicious Circuits in FPGA Bitstreams

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Elnaggar, Rana; Chaudhuri, Jayeeta; Karri, Ramesh; Chakrabarty, Krishnendu

被引用次数: 7

2023

Unsupervised Two-Stage Root-Cause Analysis With Transfer Learning for Integrated Systems

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Pan, Renjian; Li, Xin; Chakrabarty, Krishnendu

被引用次数: 0

2023

Hardware-Supported Patching of Security Bugs in Hardware IP Blocks

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Liu, Wei-Kai; Tan, Benjamin; Fung, Jason M.; Karri, Ramesh; Chakrabarty, Krishnendu

被引用次数: 2

2023

Sensor-based evaluation of a Urine Trap toilet in a shared bathroom

SCIENCE OF THE TOTAL ENVIRONMENT

Kachoria, Prateek; Sasidaran, Sarani; Welling, Claire M.; Rosario, Praveen; Zhou, Jin; Chakrabarty, Krishnendu; Grundl, Harald; Kristoferitsch, Lotte; Grego, Sonia

被引用次数: 4

2023

Characterizing Coherent Integrated Photonic Neural Networks Under Imperfections

JOURNAL OF LIGHTWAVE TECHNOLOGY

Banerjee, Sanmitra; Nikdast, Mahdi; Chakrabarty, Krishnendu

被引用次数: 9

2023

Built-In Self-Test of High-Density and Realistic ILV Layouts in Monolithic 3-D ICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chaudhuri, Arjun; Banerjee, Sanmitra; Kim, Jinwoo; Lim, Sung Kyu; Chakrabarty, Krishnendu

被引用次数: 3

2023

On the Impact of Uncertainties in Silicon-Photonic Neural Networks

IEEE DESIGN & TEST

Banerjee, Sanmitra; Nikdast, Mahdi; Chakrabarty, Krishnendu

被引用次数: 1

2023

Securing Heterogeneous 2.5D ICs Against IP Theft through Dynamic Interposer Obfuscation

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Talukdar, Jonti; Chaudhuri, Arjun; Kim, Jinwoo; Lim, Sung Kyu; Chakrabarty, Krishnendu

被引用次数: 4

2023

Unsupervised Two-Stage Root-Cause Analysis for Integrated Systems

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Pan, Renjian; Zhang, Zhaobo; Li, Xin; Chakrabarty, Krishnendu; Gu, Xinli

被引用次数: 2

2022

Knowledge Transfer in Board-Level Functional Fault Diagnosis Enabled by Domain Adaptation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Liu, Mengyun; Li, Xin; Chakrabarty, Krishnendu; Gu, Xinli

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2022

A Resilient and Hierarchical IoT-Based Solution for Stress Monitoring in Everyday Settings

IEEE INTERNET OF THINGS JOURNAL

Jiang, Shiyi; Firouzi, Farshad; Chakrabarty, Krishnendu; Elbogen, Eric B.

被引用次数: 11

2022

Functional Criticality Analysis of Structural Faults in AI Accelerators

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chaudhuri, Arjun; Talukdar, Jonti; Su, Fei; Chakrabarty, Krishnendu

被引用次数: 2

2022

Accelerating Large-Scale Graph Neural Network Training on Crossbar Diet

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ogbogu, Chukwufumnanya; Arka, Aqeeb Iqbal; Joardar, Biresh Kumar; Doppa, Janardhan Rao; Li, Hai; Chakrabarty, Krishnendu; Pande, Partha Pratim

被引用次数: 5

2022

Efficient Regulation of Synthetic Biocircuits Using Droplet-Aliquot Operations on MEDA Biochips

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ibrahim, Mohamed; Zhong, Zhanwei; Bhattacharya, Bhargab B.; Chakrabarty, Krishnendu

被引用次数: 1

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Formal Synthesis of Adaptive Droplet Routing for MEDA Biochips

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Elfar, Mahmoud; Liang, Tung-Che; Chakrabarty, Krishnendu; Pajic, Miroslav

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2022

Securing SoCs With FPGAs Against Rowhammer Attacks

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Elnaggar, Rana; Chen, Siyuan; Song, Peilin; Chakrabarty, Krishnendu

被引用次数: 2

2022

Accurate and Robust Malware Detection: Running XGBoost on Runtime Data From Performance Counters

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Elnaggar, Rana; Servadei, Lorenzo; Mathur, Shubham; Wille, Robert; Ecker, Wolfgang; Chakrabarty, Krishnendu

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2022

Efficient Identification of Critical Faults in Memristor-Based Inferencing Accelerators

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chen, Ching-Yuan; Chakrabarty, Krishnendu

被引用次数: 3

2022

C-Testing and Efficient Fault Localization for AI Accelerators

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chaudhuri, Arjun; Liu, Chunsheng; Fan, Xiaoxin; Chakrabarty, Krishnendu

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High-Throughput Training of Deep CNNs on ReRAM-Based Heterogeneous Architectures via Optimized Normalization Layers

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Joardar, Biresh Kumar; Deshwal, Aryan; Doppa, Janardhan Rao; Pande, Partha Pratim; Chakrabarty, Krishnendu

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Mixing Models as Integer Factorization: A Key to Sample Preparation With Microfluidic Biochips

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Kundu, Debraj; Roy, Sudip; Bhattacharjee, Sukanta; Saha, Sohini; Chakrabarty, Krishnendu; Chakrabarti, Partha Pratim; Bhattacharya, Bhargab B.

被引用次数: 5

2022

Runtime Malware Detection Using Embedded Trace Buffers

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Elnaggar, Rana; Basu, Kanad; Chakrabarty, Krishnendu; Karri, Ramesh

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Built-in Self-Test and Fault Localization for Inter-Layer Vias in Monolithic 3D ICs

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Chaudhuri, Arjun; Banerjee, Sanmitra; Kim, Jinwoo; Park, Heechun; Ku, Bon Woong; Kannan, Sukeshwar; Chakrabarty, Krishnendu; Lim, Sung Kyu

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2022

Design Automation and Test Solutions for Monolithic 3D ICs

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Zhu, Lingjun; Chaudhuri, Arjun; Banerjee, Sanmitra; Murali, Gauthaman; Vanna-lampikul, Pruek; Chakrabarty, Krishnendu; Lim, Sung Kyu

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Online Fault Detection in ReRAM-Based Computing Systems for Inferencing

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Liu, Mengyun; Chakrabarty, Krishnendu

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2022

Characterization and Optimization of Coherent MZI-Based Nanophotonic Neural Networks Under Fabrication Non-Uniformity

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Mirza, Asif; Shafiee, Amin; Banerjee, Sanmitra; Chakrabarty, Krishnendu; Pasricha, Sudeep; Nikdast, Mahdi

被引用次数: 11

2022

Structural Attacks and Defenses for Flow-Based Microfluidic Biochips

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Baban, Navajit Singh; Saha, Sohini; Orozaliyev, Ajymurat; Kim, Jongmin; Bhattacharjee, Sukanta; Song, Yong-Ak; Karri, Ramesh; Chakrabarty, Krishnendu

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2022

Don't CWEAT It: Toward CWE Analysis Techniques in Early Stages of Hardware Design

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Ahmad, Baleegh; Liu, Wei-Kai; Collini, Luca; Pearce, Hammond; Fung, Jason M.; Valamehr, Jonathan; Bidmeshki, Mohammad; Sapiecha, Piotr; Brown, Steve; Chakrabarty, Krishnendu; Karri, Ramesh; Tan, Benjamin

被引用次数: 5

2022

Machine Learning for Testing Machine-Learning Hardware: A Virtuous Cycle

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Chaudhuri, Arjun; Talukdar, Jonti; Chakrabarty, Krishnendu

被引用次数: 0

2022

Deep Neural Network Piration without Accuracy Loss

2022 21ST IEEE INTERNATIONAL CONFERENCE ON MACHINE LEARNING AND APPLICATIONS, ICMLA

Ray, Aritra; Jia, Jinyuan; Saha, Sohini; Chaudhuri, Jayeeta; Gong, Neil Zhenqiang; Chakrabarty, Krishnendu

被引用次数: 0

2022

NoC-enabled 3D Heterogeneous Manycore Systems for Big-Data Applications

PROCEEDINGS OF THE TWENTY THIRD INTERNATIONAL SYMPOSIUM ON QUALITY ELECTRONIC DESIGN (ISQED 2022)

Joardar, Biresh Kumar; Doppa, Janardhan Rao; Pande, Partha Pratim; Chakrabarty, Krishnendu

被引用次数: 0

2022

Fault Diagnosis for Resistive Random-Access Memory and Monolithic Inter-tier Vias in Monolithic 3D Integration

2022 IEEE INTERNATIONAL TEST CONFERENCE (ITC)

Hung, Shao-Chun; Chaudhuri, Arjun; Banerjee, Sanmitra; Chakrabarty, Krishnendu

被引用次数: 4

2022

Automatic Structural Test Generation for Analog Circuits using Neural Twins

2022 IEEE INTERNATIONAL TEST CONFERENCE (ITC)

Talukdar, Jonti; Chaudhuri, Arjun; Bhattacharya, Mayukh; Chakrabarty, Krishnendu

被引用次数: 3

2022

PriRecT: Privacy-preserving Job Recommendation Tool for GPU Sharing

2022 IEEE CLOUD SUMMIT

Ray, Aritra; Zhang, Zhaobo; Xiong, Ying; Chakrabarty, Krishnendu

被引用次数: 0

2022

Pruning Coherent Integrated Photonic Neural Networks Using the Lottery Ticket Hypothesis

2022 IEEE COMPUTER SOCIETY ANNUAL SYMPOSIUM ON VLSI (ISVLSI 2022)

Banerjee, Sanmitra; Nikdast, Mahdi; Pasricha, Sudeep; Chakrabarty, Krishnendu

被引用次数: 5

2022

Probabilistic Fault Grading for AI Accelerators using Neural Twins

2022 IEEE COMPUTER SOCIETY ANNUAL SYMPOSIUM ON VLSI (ISVLSI 2022)

Chaudhuri, Arjun; Talukdar, Jonti; Chakrabarty, Krishnendu

被引用次数: 0

2022

Fault Modeling and Efficient Testing of Memristor-Based Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Peng; You, Zhiqiang; Wu, Jigang; Liu, Bosheng; Han, Yinhe; Chakrabarty, Krishnendu

被引用次数: 17

2021

Computer-aided Design Techniques for Flow-based Microfluidic Lab-on-a-chip Systems

ACM COMPUTING SURVEYS

Huang, Xing; Ho, Tsung-Yi; Guo, Wenzhong; Li, Bing; Chakrabarty, Krishnendu; Schlichtmann, Ulf

被引用次数: 23

2021

Thwarting Bio-IP Theft Through Dummy-Valve-Based Obfuscation

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Shayan, Mohammed; Bhattacharjee, Sukanta; Orozaliyev, Ajymurat; Song, Yong-Ak; Chakrabarty, Krishnendu; Karri, Ramesh

被引用次数: 8

2021

Enhancing the Reliability of MEDA Biochips Using JTAG and Wear Leveling

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhong, Zhanwei; Liang, Tung-Che; Chakrabarty, Krishnendu

被引用次数: 5

2021

Board-Level Functional Fault Identification Using Streaming Data

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Liu, Mengyun; Ye, Fangming; Li, Xin; Chakrabarty, Krishnendu; Gu, Xinli	2021
被引用次数: 7	
Security Against Data-Sniffing and Alteration Attacks in IJTAG	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Elnaggar, Rana; Karri, Ramesh; Chakrabarty, Krishnendu	
被引用次数: 4	2021
Toward Hardware-Based IP Vulnerability Detection and Post-Deployment Patching in Systems-on-Chip	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Tan, Benjamin; Elnaggar, Rana; Fung, Jason M.; Karri, Ramesh; Chakrabarty, Krishnendu	
被引用次数: 10	2021
AccuReD: High Accuracy Training of CNNs on ReRAM/GPU Heterogeneous 3-D Architecture	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Joardar, Biresh Kumar; Doppa, Janardhan Rao; Pande, Partha Pratim; Li, Hai; Chakrabarty, Krishnendu	
被引用次数: 31	2021
Black-Box Test-Cost Reduction Based on Bayesian Network Models	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Pan, Renjian; Zhang, Zhaobo; Li, Xin; Chakrabarty, Krishnendu; Gu, Xinli	
被引用次数: 12	2021
How Secure Are Checkpoint-Based Defenses in Digital Microfluidic Biochips?	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Shayan, Mohammed; Bhattacharjee, Sukanta; Wille, Robert; Chakrabarty, Krishnendu; Karri, Ramesh	
被引用次数: 8	2021
Access-Time Minimization for the IJTAG Network Using Data Broadcast and Hardware Parallelism	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Zhong, Zhanwei; Li, Guoliang; Yang, Qinfu; Chakrabarty, Krishnendu	
被引用次数: 5	2021
Learning to Train CNNs on Faulty ReRAM-based Manycore Accelerators	
ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS	
Joardar, Biresh Kumar; Doppa, Janardhan Rao; Li, Hai; Chakrabarty, Krishnendu; Pande, Partha Pratim	
被引用次数: 12	2021
Lotus: A New Topology for Large-scale Distributed Machine Learning	
ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS	
Lu, Yunfeng; Gu, Huaxi; Yu, Xiaoshan; Chakrabarty, Krishnendu	
被引用次数: 2	2021
Power Supply Noise-Aware At-Speed Delay Fault Testing of Monolithic 3-D ICs	

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hung, Shao-Chun; Lu, Yi-Chen; Lim, Sung Kyu; Chakrabarty, Krishnendu

被引用次数: 2

2021

Performance and Accuracy Tradeoffs for Training Graph Neural Networks on ReRAM-Based Architectures

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Arka, Aqeeb Iqbal; Joardar, Biresh Kumar; Doppa, Janardhan Rao; Pande, Partha Pratim; Chakrabarty, Krishnendu

被引用次数: 17

2021

Variation-Aware Delay Fault Testing for Carbon-Nanotube FET Circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Banerjee, Sanmitra; Chaudhuri, Arjun; Ning, August; Chakrabarty, Krishnendu

被引用次数: 10

2021

Acoustohydrodynamic tweezers via spatial arrangement of streaming vortices

SCIENCE ADVANCES

Zhu, Haodong; Zhang, Peiran; Zhong, Zhanwei; Xia, Jianping; Rich, Joseph; Mai, John; Su, Xingyu; Tian, Zhenhua; Bachman, Hunter; Rufo, Joseph; Gu, Yuyang; Kang, Putong; Chakrabarty, Krishnendu; Witelski, Thomas P.; Huang, Tony Jun

被引用次数: 40

2021

Acoustoelectronic nanotweezers enable dynamic and large-scale control of nanomaterials

NATURE COMMUNICATIONS

Zhang, Peiran; Rufo, Joseph; Chen, Chuyi; Xia, Jianping; Tian, Zhenhua; Zhang, Liying; Hao, Nanjing; Zhong, Zhanwei; Gu, Yuyang; Chakrabarty, Krishnendu; Huang, Tony Jun

被引用次数: 42

2021

Parallel Droplet Control in MEDA Biochips using Multi-Agent Reinforcement Learning

International Conference on Machine Learning

Tung-Che Liang; Jin Zhou; Yun-Sheng Chan; Tsung-Yi Ho; Krishnendu Chakrabarty; Cy Lee

被引用次数: 0

2021

Pruning of Deep Neural Networks for Fault-Tolerant Memristor-based Accelerators

Design Automation Conference

Ching-Yuan Chen; Krishnendu Chakrabarty

被引用次数: 0

2021

Testing and Fault-Localization Solutions for Monolithic 3D ICs

2021 IEEE INTERNATIONAL TEST CONFERENCE IN ASIA (ITC-ASIA 2021)

Chaudhuri, Arjun; Chakrabarty, Krishnendu

被引用次数: 1

2021

Securing Biochemical Samples Using Molecular Barcoding on Digital Microfluidic Biochips

2021 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE (IEEE BIOCAS 2021)

Liang, Tung-Che; Abaffy, Tatjana; Matsunami, Hiroaki; Chakrabarty, Krishnendu; Karri, Ramesh

被引用次数: 3	2021
Modeling Silicon-Photonic Neural Networks under Uncertainties	
PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)	
Banerjee, Sanmitra; Nikdast, Mandi; Chakrabarty, Krishnendu	
被引用次数: 18	2021
Advances in Testing and Design-for-Test Solutions for M3D Integrated Circuits	
PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)	
Banerjee, Sanmitra; Chaudhuri, Arjun; Hung, Shan-Chun; Chakrabarty, Krishnendu	
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Formal Synthesis of Adaptive Droplet Routing for MEDA Biochips	
PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)	
Elfar, Mahmoud; Liang, Tung-Che; Chakrabarty, Krishnendu; Pajic, Miroslav	
被引用次数: 4	2021
Efficient Identification of Critical Faults in Memristor Crossbars for Deep Neural Networks	
PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)	
Chen, Ching-Yuan; Chakrabarty, Krishnendu	
被引用次数: 7	2021
Fault-Criticality Assessment for AI Accelerators using Graph Convolutional Networks	
PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)	
Chaudhuri, Arjun; Talukdar, Jonti; Jung, Jinwook; Nam, Gi-Joon; Chakrabarty, Krishnendu	
被引用次数: 6	2021
ReGraphX: NoC-enabled 3D Heterogeneous ReRAM Architecture for Training Graph Neural Networks	
PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)	
Arka, Aqeeb Iqbal; Joardar, Biresh Kumar; Doppa, Janardhan Rao; Pande, Partha Pratim; Chakrabarty, Krishnendu	
被引用次数: 27	2021
Efficient Fault-Criticality Analysis for AI Accelerators using a Neural Twin	
2021 IEEE INTERNATIONAL TEST CONFERENCE (ITC 2021)	
Chaudhuri, Arjun; Chen, Ching-Yuan; Talukdar, Jonti; Madala, Siddarth; Dubey, Abhishek Kumar; Chakrabarty, Krishnendu	
被引用次数: 12	2021
On-line Functional Testing of Memristor-mapped Deep Neural Networks using Backdoored Checksums	

2021 IEEE INTERNATIONAL TEST CONFERENCE (ITC 2021)

Chen, Ching-Yuan; Chakrabarty, Krishnendu

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2021

Adaptive Methods for Machine Learning-Based Testing of Integrated Circuits and Boards

2021 IEEE INTERNATIONAL TEST CONFERENCE (ITC 2021)

Liu, Mengyun; Chakrabarty, Krishnendu

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2021

A BIST-based Dynamic Obfuscation Scheme for Resilience against Removal and Oracle-guided Attacks

2021 IEEE INTERNATIONAL TEST CONFERENCE (ITC 2021)

Talukdar, Jonti; Chen, Siyuan; Das, Amitabh; Aftabjahani, Sohrab; Song, Peilin; Chakrabarty, Krishnendu

被引用次数: 3

2021

DARe: DropLayer-Aware Manycore ReRAM architecture for Training Graph Neural Networks

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Arka, Aqeeb Iqbal; Joardar, Biresh Kumar; Doppa, Janardhan Rao; Pande, Partha Pratim; Chakrabarty, Krishnendu

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2021

ParaMitE: Mitigating Parasitic CNFETs in the Presence of Unetched CNTs

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Banerjee, Sanmitra; Chaudhuri, Arjun; Kim, Jinwoo; Murali, Gauthaman; Nelson, Mark; Lim, Sung Kyu; Chakrabarty, Krishnendu

被引用次数: 3

2021

Heterogeneous Manycore Architectures Enabled by Processing-in-Memory for Deep Learning: From CNNs to GNNs

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Joardar, Biresh Kumar; Arka, Aqeeb Iqbal; Doppa, Janardhan Rao; Pande, Partha Pratim; Li, Hai; Chakrabarty, Krishnendu

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Optimization of Multi-Target Sample Preparation On-Demand With Digital Microfluidic Biochips

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Toward Secure Microfluidic Fully Programmable Valve Array Biochips

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Micro-Electro-Dot-Array Digital Microfluidic Biochips: Technology, Design Automation, and Test Techniques

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Efficient and Adaptive Error Recovery in a Micro-Electro-Dot-Array Digital Microfluidic Biochip

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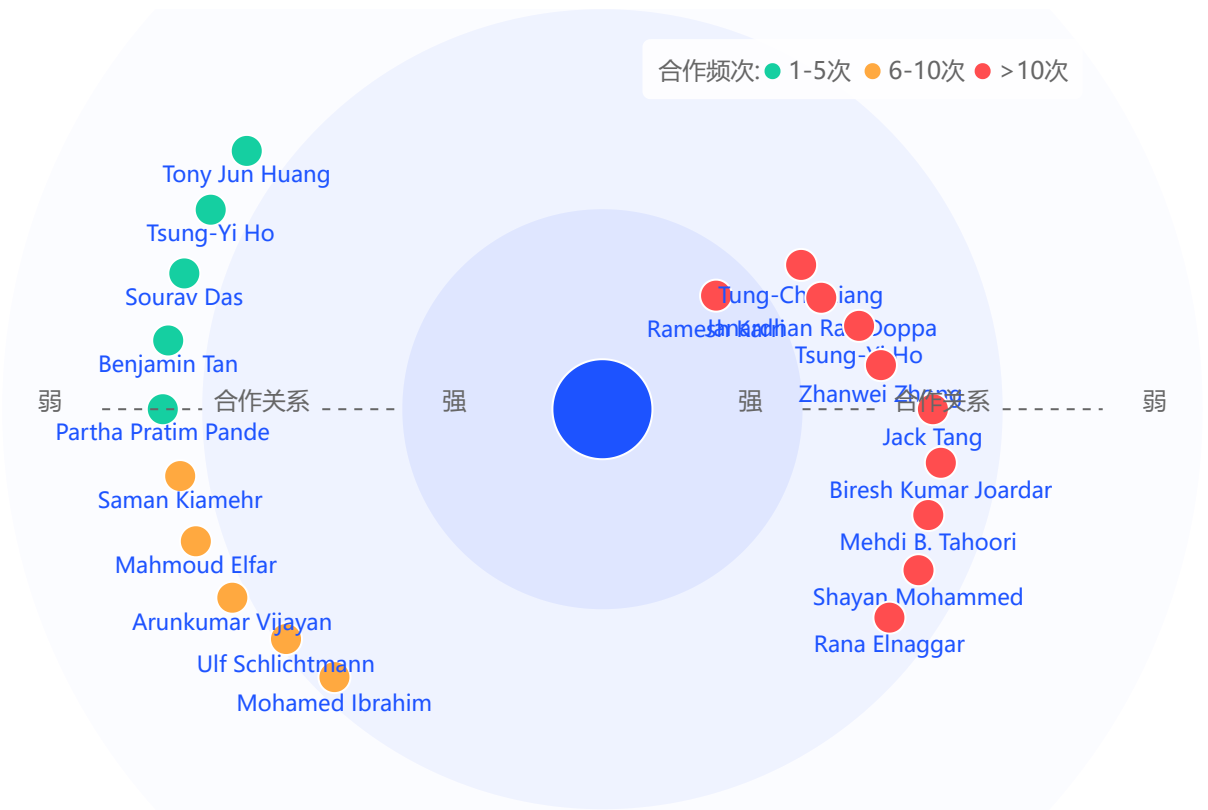
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